

Course Syllabus

2024 Fall

Course	統計諮詢專題 (Statistical Consultings)	Instructor	Dr. Mi-Chia Ma Phone: (06)2757575 ext. 53639 Office: 62439(4 th Fl.) Dept. of Statistics Email:mcma@ncku.edu.tw
Objectives	1. Review the statistical analysis methods 2. Communicate with clients 3. Apply the statistical methods and package to complete the analysis report of real data		
Class	Institute of Statistics	Level	Master
Credits	3 hours/week	Prerequisite	Multivariate Analysis, Experimental Design, Regression Analysis, Time series
Contents	1. 實驗設計; 模糊理論 2. 類別資料分析(含multi-logit model); 非線性迴歸分析 3. 關聯規則&支持向量機器方法(SVM); 決策樹&隨機森林&類神經網路 4. 主成分分析&因素分析; 結構方程模型(SEM) 5. 區別分析&群集分析; 多元尺度分析&典型相關分析 6. 存活分析(Kaplan Meier Estimator, Log Rank test); Cox regression Model 7. 重複測量&多變量變異數分析(MANOVA)&廣義估計方程(GEE) 8. 時間序列(AR(P), MA(q)); ARIMA(p, q) 9. 對應分析(correspondence); 分析階層方法(AHP); 資料包絡分析法(data envelopment analysis, DEA)		
Reference	1. Heiberger R.M. and Holland, B. (2015) Statistical Analysis and Data Display-An Intermediate Course with Examples in R, Springer. 2. Applied Linear Regression Models by Neter et al. 3. Applied Nonparameter Statistics by Daniel. 4. Applied Multivariate Statistical Analysis by Johnson & Wichern.		
Grading policy	1. Midterm Report (期中個人上台報告+書面報告) 30% 2. Final Report (期末分組上台報告+書面報告) 30% 3. Homework and Class Participation (作業及出席狀況) 40%		